

RAMCO INSTITUTE OF TECHNOLOGY

RAJAPALAYAM

DEPT. OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INDUSTRIAL VISIT MEETING MINUTES

05.03.2024

Date & Venue: 01.03.2024 & Ramco Cements, Ariyalur

Agenda: Opportunities of Artificial Intelligence tools in cement Industry

Attendees

Academicians

SNo	Name	Designation	Department
1	Dr.M.Kaliappan	Professor	AI & DS
2	Harshavardhini A S	Student- III Year	AI & DS
3	Iniyan K	Student- III Year	AI & DS
4	Mareeswaran M	Student- III Year	AI & DS
5	Arumuga Tamil selvan T	Student- II Year	AI & DS
6	Saravana Kumar M	Student- II Year	AI & DS
7	Dharnees V	Student- II Year	EEE
8	Muthu Selvam S	Student- II Year	EEE

Industrialist

1. Mr.Gulkarni, Vice President(works), Ramco Cements, Aerialur
2. Mr.A.K Ganeshram, Senior General Manager, E&I, Ramco Cements, Aerialur
3. Unit Head, IT, Ramco Cements, Aerialur
4. Unit Head, Production, Ramco Cements, Aerialur
5. Unit Head, Sales & Inventory, Ramco Cements, Aerialur
6. Unit Head, Energy, Ramco Cements, Aerialur
7. Unit Head, Process control, Ramco Cements, Aerialur
8. Unit Head, Quality control, Ramco Cements, Aerialur

Discussions:

- Dr.M.Kaliappan, Professor, Ramco Institute of Technology, introduced the RIT and academicians such 5 students from AI & DS and 2 Students from EEE department
- He presented the various opportunities of AI tools in the various cement industry such as production, inventory, sales and electrical and instrumentation department and how AI tools can be utilized to address them.
- All unit heads are expressed their real time problem statement raised in the cement industry and scope of the AI tools in that area.
- The following are discussed by the all unit heads with academicians
 - Process Optimization
 - Quality Control - [Include Data's in Parameters]
 - Energy Loss Managements

- Interventions Minimization
- Sales Demand Forecasting [Include Data's - Credential]
- Payment Processing Automation Software
- Parking Slot Vacant and ideal vehicle Detection
- Face Recognition [Stranger's entry monitoring, Truck Entities - (CV Number Plate Detection)]
- Ariyalur Weather Forecasting on the mining Area using AI
- Safety Precautions - Helmet & Shoes Detection in the Main Blocks of Factory
- Bag weight Defect adjustment using AI [Hard, Data Include for Each Pack]
- Carbon Absorbing of Tree's Calculation [BI- Dashboard, Data include]
- Auto Compression in Boilers & Turbine with vibrational sensor & Parameters monitoring
- Cement Bag count detection and correction



Conclusion:

RIT would like to provide a structured and systematic approach to solving problems using interdisciplinary teams. By sending interns from different department such as Artificial Intelligence & Data Science, Electrical & Electronics Engineering, and Mechanical Engineering, RIT aims to address various types of problems with diverse perspectives and expertise. Supervision from Dr. M. Kaliappan and Dr. S. Vimal ensures that the interns have guidance and support throughout their internship experience. This approach fosters collaboration, innovation, and practical application of knowledge, ultimately leading to effective problem-solving solutions.

Prepared by

Dr.M. Kaliappan, Professor/AD